



Exam :

End Term Quiz

Subject :

Maths2

Total Marks :

50.00

QP :

2023 Dec24: IIT M FN EXAM FDI2

Exam Mode

Learning Mode

View Question Paper Summary

QUESTION MENU

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CONTROLS

SUBMIT EXAM

Your Score

0.00 / 50.00

(0%)

Question 1 : 640653701165

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "FOUNDATION LEVEL : MATHEMATICS FOR DATA SCIENCE II (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

☐ YES

☐ NO

Your score : 0

 Discussions (0)



Question 2 : 640653701166

 View Solutions (0)

Total Mark : 3.00 | Type : MCQ

Choose the correct statement.

OPTIONS :

- ☐ If A and B are two matrices such that both AB and BA are well defined, then both A and B are square matrices. 
- ☐ If A is a square matrix of order 2 such that $A^2 = I$, then A has to be equal to I . 
- ☐ A and B are square matrices of order n . If B is not invertible, then AB is not invertible. 
- ☐ A and B are square matrices of order n and $k \leq n$. If $\text{rank}(AB) = \text{rank}(B) = k$, then $\text{rank}(A) < k$. 

Your score : 0

 Discussions (0)



Question 3 : 640653701176

 View Solutions (0)

Total Mark : 3.00 | Type : MCQ

Let V be a vector space and v_1, v_2, v_3 be three distinct elements in V . If every $v \in V$ can be expressed as a *unique* linear combination of these three vectors, which of the following statements is true? Here, $\dim(V)$ refers to the dimension of V .

OPTIONS :

- ☐ $\dim(V) = 3$
- ☐ We cannot comment on the exact value of $\dim(V)$. It lies in this range:
 $1 \leq \dim(V) \leq 3$
- ☐ We cannot comment on the exact value of $\dim(V)$. It lies in this range:
 $0 \leq \dim(V) \leq 3$
- ☐ We cannot pass any judgement whatsoever on the value of $\dim(V)$ with the given data.

Your score : 0

Discussions (0)



Question 4 : 640653701171

View Solutions (0)

Total Mark : 4.00 | Type : MCQ

Consider the following system:

$$x + 2y - 2z = b_1$$

$$2x + 5y - 4z = b_2$$

$$4x + 9y - 8z = b_3$$

What conditions should b_1, b_2, b_3 satisfy so that this system has at least one solution?

OPTIONS :

- ☐ $2b_1 + b_2 - b_3 = 0$
- ☐ $b_2 = 2b_1$
- ☐ $b_3 = 4b_1$
- ☐ This system always has a solution for any set of values for b_1, b_2, b_3

☐ This system doesn't have a solution for any set of values of b_1, b_2, b_3 . 

Your score : 0

 Discussions (0)



Question 5 : 640653701185

 View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function given by 

$$f(x, y) = \begin{cases} \frac{(x+y)^2}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Choose the correct option(s) from the following:

OPTIONS :

☐ $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists along the positive x -axis. 

☐ $f(x, y)$ approaches 0 when (x, y) approaches $(0, 0)$ along the positive y -axis. 

☐ $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist. 

☐ $f_x(0, 0) = 1$. 

☐ The directional derivative of f at $(0, 0)$ in the direction of positive x -axis does not exist. 

☐ The directional derivative of f at $(0, 0)$ in the direction of positive y -axis exists. 

Your score : 0

 Discussions (0)



Question 6 : 640653701194

[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : MSQ

Let $f(x, y) = x^3 + 2x^2y + y^2 - 2x + y + 1$. Choose the correct option(s) from the following:

OPTIONS :

- ☐ The tangent plane of f at $(1, -1)$ is given by $z = 3x - y - 2$.
- ☐ The tangent plane of f at $(1, -1)$ is given by $3x - y + z - 2 = 0$.
- ☐ The best linear approximation for f close to the point $(1, 1)$ is given by $L_f(x, y) = 5x + 5y - 6$.
- ☐ The best linear approximation for f close to the point $(1, 1)$ is given by $L_f(x, y) = x - 5y + 6$.
- ☐ If T_1 is the tangent plane of f at $(1, -1)$ and T_2 is the tangent plane of f at $(1, 1)$, then T_1 and T_2 intersect at infinitely many points.
- ☐ If T_1 is the tangent plane of f at $(1, -1)$ and T_2 is the tangent plane of f at $(1, 1)$, then T_1 and T_2 intersect at only one point.

Your score : 0

[Discussions \(0\)](#)

Question 7 : 640653701195

[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : MSQ

Let $f(x, y) = x^2 - y^2 + 4$ on the disc $S = \{(x, y) : x^2 + y^2 \leq 1\}$. Choose the correct option(s) from the following:

OPTIONS :

- ☐ The absolute maximum for f occurs at two points and is equal to 5.
- ☐ The absolute maximum for f occurs only at the point $(1, 0)$.
- ☐ The absolute minimum for f occurs only at $(0, 1)$.

- ☐ The absolute minimum for f occurs at two points on the boundary of the disc S .
- ☐ $(0, 0)$ is a saddle point of f .
- ☐ There are no saddle points for f .

Your score : 0

Discussions (0)



Question 8 : 640653701170

View Solutions (0)

Total Mark : 2.00 | Type : SA

A is a square matrix of order 5. When the following sequence of row operations are performed on A , the result is the identity matrix:

- Swap the second and fifth rows.
- Add three times the fourth row to the first row.
- Divide the third row by 4.
- Multiply the first row by 2.

Compute $\det(A)$.

Answer (Numeric):

Answer

Accepted Answer : -2

Your score : 0

Discussions (0)



Question 9 : 640653701167

Total Mark : 0.00 | Type : COMPREHENSION

Consider the subspace $W = \{(x, y, z, w) \mid x + z = y, y + w = z - x\}$ of $\mathbb{R}^4$.

Based on the above data, answer the given subquestions.

Your score : 0

**Question 10 :**
640653701168

View Parent QN



View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Which of the following are bases for W ?
Select all correct options.

OPTIONS :

☐ $\beta = \{(-1, -1, 0, 2)\}$

☐ $\beta = \{(-1, -1, 0, 2), (0, 1, 1, 0)\}$

☐ $\beta = \{(1, 1, 0, -2), (1, 0, -1, -2)\}$

☐ $\beta = \{(1, 1, 0, -2), (0, 1, 1, 0), (2, 1, -1, -4)\}$

☐ $\beta = \{(-1, -1, 0, 2), (2, 1, -1, -4), (0, 1, 1, 0)\}$

Your score : 0

Discussions (0)

**Question 11 :**
640653701169

View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

What is the dimension of W ?

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)



Question 12 : 640653701177

Total Mark : 0.00 | Type : COMPREHENSION

Suppose $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is an isomorphism. Let A be the matrix representation of T with respect to the standard ordered basis $\mathcal{B} = \{e_1, e_2, e_3\}$ for both domain and co-domain and B be the matrix representation of T with respect to the basis $\mathcal{B}' = \{e_1, e_1 - e_2, e_2 - e_3\}$ for both domain and co-domain.

Based on the above data, answer the given subquestions.

Your score : 0

Question 13 :
640653701178

View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Find $\text{Rank}(A)$.

Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

Discussions (0)

Question 14 :
640653701179

View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the dimension of the solution space of the homogeneous equation $Bx = 0$.

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)



Question 15 :
640653701180

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Choose the correct option(s) from the following:

OPTIONS :

☐ A^{-1} and B^{-1} are equivalent but not similar matrices.

☐ The matrix A is invertible but B is not invertible.

☐ A^{-1} and B^{-1} are similar matrices.

☐ AB is similar to BA .

Your score : 0

Discussions (0)



Question 16 : 640653701181

Total Mark : 0.00 | Type : COMPREHENSION

Let $W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y = z\}$ be a subspace of the inner product space $\mathbb{R}^3$ with respect to the dot product.



Based on the above data, answer the given subquestions.

Your score : 0



Question 17 :
640653701182



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Find $\dim(W^\perp)$.



Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

Discussions (0)



Question 18 :
640653701183



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

If (a, b, c) is a vector in $W^\perp$,
then $2a + b$ equals



Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)



Question 19 :

640653701184

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Choose the correct option(s) from the following:

OPTIONS :

- ☐ If P_W represents the projection from $\mathbb{R}^3$ to W , then $W^\perp$ is the nullspace of P_W .
- ☐ The range of P_W is given by $\text{span}\{(1, 0, 1), (0, 1, 2)\}$.
- ☐ If $v \in \text{span}\{(1, 0, 1), (0, 1, -2)\}$, then $(I - P_w)(v) = 0$ where I is the identity transformation on $\mathbb{R}^3$.
- ☐ If S is a subspace of W , then $S^\perp \subseteq W^\perp$.

Your score : 0

Discussions (0)



Question 20 : 640653701172

Total Mark : 0.00 | Type : COMPREHENSION

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined as follows:

$$T(x, y, z) = (x + z, z - y, x - y + 2z)$$

Based on the above data, answer the given subquestions.

Your score : 0



Question 21 :
640653701173

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

Find the nullity of T . 

Answer (Numeric):

Accepted Answer : 1

Your score : 0

[Discussions \(0\)](#)

Question 22 :
640653701174

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

Find the rank of T . 

Answer (Numeric):

Accepted Answer : 2

Your score : 0

[Discussions \(0\)](#)

Question 23 :
640653701175

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : MCQ

Choose the correct option.

OPTIONS :

☐ T is an isomorphism. 

- ☐ T is one-one but not onto. 
- ☐ T is onto but not one-one. 
- ☐ T is neither one-one nor onto. 

Your score : 0

 Discussions (0)



Question 24 : 640653701186

Total Mark : 0.00 | Type : COMPREHENSION

$$\text{Let } f(x, y) = \begin{cases} \frac{\cos y \sin x}{x} & x \neq 0 \\ \cos y & x = 0 \end{cases}.$$



Based on the above data, answer the given subquestions.

Your score : 0



Question 25 : 640653701187

 View Parent QN

 View Solutions (0)

Total Mark : 1.00 | Type : SA

Find $\lim_{(x,y) \rightarrow (0,0)} f(x, y).$



Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

[Discussions \(0\)](#)

Question 26 :
640653701188

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

Is the function continuous at $(0,0)$? 

[If the answer is yes, enter 1 and
if the answer is no, enter 0.]

Answer (Numeric):

Accepted Answer : 1

Your score : 0

[Discussions \(0\)](#)

Question 27 :
640653701189

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

At how many points does f have
a discontinuity? 

Answer (Numeric):

Accepted Answer : 0

Your score : 0

[Discussions \(0\)](#)

Question 28 : 640653701190

Total Mark : 0.00 | Type : COMPREHENSION

Let $f(x, y) = 2x^2 + \frac{3y^2}{2}$. If $u = (a, b)$ is the direction in which



Based on the above data, answer the given subquestions.

Your score : 0



Question 29 :
640653701191



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

f increases most rapidly at the point $(1, 1)$, find $5(a + b)$.



Answer (Numeric):

Answer

Accepted Answer : 7

Your score : 0

Discussions (0)



Question 30 :
640653701192



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

f decreases most rapidly at the point $(1, 1)$, find $5(a + b)$.



Answer (Numeric):

Answer

Accepted Answer : -7

Your score : 0

[Discussions \(0\)](#)

Question 31 :
640653701193

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

there is no change in f at the point $(1, 1)$, find $5|a + b|$.



Answer (Numeric):

Accepted Answer : 1

Your score : 0

[Discussions \(0\)](#)

Question 32 : 640653701196

Total Mark : 0.00 | Type : COMPREHENSION

Let $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 5$. Use the Hessian test to answer the given subquestions.



Your score : 0



Question 33 :
640653701197

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

Find the number of local maxima.

Answer (Numeric):

Accepted Answer : 1

Your score : 0

 Discussions (0)



Question 34 :
640653701198



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : SA

Find the number of local minima.

Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

 Discussions (0)



Question 35 :
640653701199



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : SA

Find the number of saddle points.

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

 Discussions (0)



Question 36 :
640653701200



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

If there are no local maxima, enter the answer as 100. Else, let (a, b) be a local maximum such that it is farthest from the origin, and if there are more than one such points, then it has the largest x -coordinate amongst them. Find $f(a, b)$.



Answer (Numeric):

Answer

Accepted Answer : 5

Your score : 0

Discussions (0)



Question 37 :
640653701201



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

If there are no local minima, enter the answer as 100. Else, let (a, b) be a local minimum such that it is farthest from the origin, and if there are more than one such points, then it has the largest x -coordinate amongst them. Find $f(a, b)$.



Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

Discussions (0)



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